

SQL Task:1

Question 1 and 2

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MySQL Shell 8.0.46

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Type '\help' or '?' for help; '\quit' to exit.
MySQL_JS> \sql
Switching to SQL mode... Commands end with ;
MySQL_SQL> \c root@localhost
Creating a session to 'root@localhost'
Fetching global names for auto-completion... Press ^C to stop.
Your MySQL connection id is 22 (X protocol)
Server version: 8.0.46 MySQL Community Server - GPL
No default schema selected; type \use <schema> to set one.
MySQL localhost:33060+ ssl SQL> create database employee6;
Query OK, 1 row affected (0.0069 sec)
MySQL localhost:33060+ ssl SQL> use employee6;
Default schema set to 'employee6'.
Fetching global names, object names from 'employee6' for auto-completion... Press ^C to stop.
MySQL localhost:33060+ ssl employee6 SQL> create table employee6(emp_id int,name varchar(20),dept varchar(20),salary bigint);
Query OK, 0 rows affected (0.0356 sec)
MySQL localhost:33060+ ssl employee6 SQL> alter table employee6 add joining_date date first;
Query OK, 0 rows affected (0.0474 sec)

Records: 0 Duplicates: 0 Warnings: 0
MySQL localhost:33060+ ssl employee6 SQL> alter table employee6 add email varchar(20) after name;
Query OK, 0 rows affected (0.0461 sec)

Records: 0 Duplicates: 0 Warnings: 0
MySQL localhost:33060+ ssl employee6 SQL> alter table employee6 rename column dept to emp_dept;
Query OK, 0 rows affected (0.0301 sec)

MySQL localhost:33060+ ssl employee6 SQL> alter table employee6 modify salary decimal(10,2);
Query OK, 0 rows affected (0.0675 sec)

Records: 0 Duplicates: 0 Warnings: 0
MySQL localhost:33060+ ssl employee6 SQL> desc employee;
ERROR: 1146: Table 'employee6.employee' doesn't exist
MySQL localhost:33060+ ssl employee6 SQL> desc employee6;
+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| joining_date | date | YES | | NULL | |
| emp_id | int | YES | | NULL | |
| name | varchar(20) | YES | | NULL | |
| email | varchar(20) | YES | | NULL | |
| emp_dept | varchar(20) | YES | | NULL | |
| salary | decimal(10,2) | YES | | NULL | |
+-----+-----+-----+-----+-----+-----+
6 rows in set (0.0029 sec)
MySQL localhost:33060+ ssl employee6 SQL> INSERT INTO employee6(joining_date, emp_id, name, email, emp_dept, salary) VALUES
-> ('2023-03-11', 111, 'Chiranjeevi K', 'chiru@company.com', 'IT', 95000.00),
-> ('2023-04-18', 112, 'Mahesh Babu', 'mahesh@company.com', 'Marketing', 88000.00),
-> ('2023-07-25', 113, 'Bhavana Reddy', 'bhavana@company.com', 'HR', 52000.50),
-> ('2023-11-02', 114, 'Satish Kumar', 'satish@company.com', 'Finance', 71000.00),
-> ('2024-02-14', 115, 'Sravani Naidu', 'sravani@company.com', 'Sales', 48000.25),
-> ('2024-05-19', 116, 'Pawan Kalyan', 'pawan@company.com', 'IT', 90000.00),
-> ('2024-08-08', 117, 'Ramakrishna M', 'ramki@company.com', 'Finance', 76000.75),
-> ('2024-12-21', 118, 'Divya Teja', 'divya@company.com', 'HR', 54000.00),
-> ('2025-03-03', 119, 'Nagarjuna Rao', 'nag@company.com', 'IT', 83000.00),
-> ('2025-04-15', 120, 'Sandhya Rani', 'sandhya@company.com', 'Marketing', 62500.50);
Query OK, 10 rows affected (0.0073 sec)
```

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MySQL Shell
Records: 10 Duplicates: 0 Warnings: 0
MySQL localhost:33060+ ssl employee6 SQL> select * from employee6;

+-----+-----+-----+-----+-----+-----+
| joining_date | emp_id | name       | email                | emp_dept | salary |
+-----+-----+-----+-----+-----+-----+
| 2023-03-11   | 111    | Chiranjeevi K | chiru@company.com    | IT       | 95000.00 |
| 2023-04-18   | 112    | Mahesh Babu   | mahesh@company.com   | Marketing | 88000.00 |
| 2023-07-25   | 113    | Bhavana Reddy | bhavana@company.com  | HR       | 52000.50 |
| 2023-11-02   | 114    | Satish Kumar  | satish@company.com   | Finance  | 71000.00 |
| 2024-02-14   | 115    | Sravani Naidu | sravani@company.com  | Sales    | 48000.25 |
| 2024-05-19   | 116    | Pawan Kalyan | pawan@company.com    | IT       | 90000.00 |
| 2024-08-08   | 117    | Ramakrishna M | ramki@company.com    | Finance  | 76000.75 |
| 2024-12-21   | 118    | Divya Teja   | divya@company.com    | HR       | 54000.00 |
| 2025-03-03   | 119    | Nagarjuna Rao | nag@company.com      | IT       | 83000.00 |
| 2025-04-15   | 120    | Sandhya Rani | sandhya@company.com  | Marketing | 62500.50 |
+-----+-----+-----+-----+-----+-----+

10 rows in set (0.0011 sec)
MySQL localhost:33060+ ssl employee6 SQL> select * from employee6 where salary>50000;

+-----+-----+-----+-----+-----+-----+
| joining_date | emp_id | name       | email                | emp_dept | salary |
+-----+-----+-----+-----+-----+-----+
| 2023-03-11   | 111    | Chiranjeevi K | chiru@company.com    | IT       | 95000.00 |
| 2023-04-18   | 112    | Mahesh Babu   | mahesh@company.com   | Marketing | 88000.00 |
| 2023-07-25   | 113    | Bhavana Reddy | bhavana@company.com  | HR       | 52000.50 |
| 2023-11-02   | 114    | Satish Kumar  | satish@company.com   | Finance  | 71000.00 |
| 2024-05-19   | 116    | Pawan Kalyan | pawan@company.com    | IT       | 90000.00 |
| 2024-08-08   | 117    | Ramakrishna M | ramki@company.com    | Finance  | 76000.75 |
| 2024-12-21   | 118    | Divya Teja   | divya@company.com    | HR       | 54000.00 |
| 2025-03-03   | 119    | Nagarjuna Rao | nag@company.com      | IT       | 83000.00 |
| 2025-04-15   | 120    | Sandhya Rani | sandhya@company.com  | Marketing | 62500.50 |
+-----+-----+-----+-----+-----+-----+

9 rows in set (0.0006 sec)
MySQL localhost:33060+ ssl employee6 SQL> select * from employee6 order by salary desc;

+-----+-----+-----+-----+-----+-----+
| joining_date | emp_id | name       | email                | emp_dept | salary |
+-----+-----+-----+-----+-----+-----+
| 2023-03-11   | 111    | Chiranjeevi K | chiru@company.com    | IT       | 95000.00 |
| 2024-05-19   | 116    | Pawan Kalyan | pawan@company.com    | IT       | 90000.00 |
| 2023-04-18   | 112    | Mahesh Babu   | mahesh@company.com   | Marketing | 88000.00 |
| 2025-03-03   | 119    | Nagarjuna Rao | nag@company.com      | IT       | 83000.00 |
| 2024-08-08   | 117    | Ramakrishna M | ramki@company.com    | Finance  | 76000.75 |
| 2023-11-02   | 114    | Satish Kumar  | satish@company.com   | Finance  | 71000.00 |
| 2025-04-15   | 120    | Sandhya Rani | sandhya@company.com  | Marketing | 62500.50 |
| 2024-12-21   | 118    | Divya Teja   | divya@company.com    | HR       | 54000.00 |
| 2023-07-25   | 113    | Bhavana Reddy | bhavana@company.com  | HR       | 52000.50 |
| 2024-02-14   | 115    | Sravani Naidu | sravani@company.com  | Sales    | 48000.25 |
+-----+-----+-----+-----+-----+-----+

10 rows in set (0.0011 sec)
MySQL localhost:33060+ ssl employee6 SQL> create table student(std_id int primary key,std_name varchar(20),course_id int);
Query OK, 0 rows affected (0.0278 sec)
MySQL localhost:33060+ ssl employee6 SQL> create table course(course_id int primary key, course_name varchar(20));
Query OK, 0 rows affected (0.0242 sec)
MySQL localhost:33060+ ssl employee6 SQL> desc student;
ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'desc student' at line 1
MySQL localhost:33060+ ssl employee6 SQL> desc student;

+-----+-----+-----+-----+-----+-----+
| Field | Type | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| std_id | int  | NO   | PRI | NULL    |       |
| std_name | varchar(20) | YES |     | NULL    |       |
| course_id | int  | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+

3 rows in set (0.0027 sec)
MySQL localhost:33060+ ssl employee6 SQL> desc course;
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MySQL Shell
3 rows in set (0.0027 sec)
MySQL localhost:33060+ ssl employee6 SQL> desc course;
+-----+-----+-----+-----+-----+-----+
| Field      | Type      | Null | Key | Default | Extra |
+-----+-----+-----+-----+-----+-----+
| course_id  | int       | NO   | PRI | NULL    |       |
| course_name | varchar(20) | YES  |     | NULL    |       |
+-----+-----+-----+-----+-----+-----+

2 rows in set (0.0032 sec)
MySQL localhost:33060+ ssl employee6 SQL> insert into student values(1,"gandadhar",101),(2,"mahesh",102),(3,"abhiram",101),(4,"s
aijahanav",102),(5,"adhi",102),(6,"lokes",102),(7,"ganesh",101),(8,"teja",102);
Query OK, 8 rows affected (0.0151 sec)

Records: 8 Duplicates: 0 Warnings: 0
MySQL localhost:33060+ ssl employee6 SQL> insert into course values(101,"python"),(102,"sql");
Query OK, 2 rows affected (0.0157 sec)

Records: 2 Duplicates: 0 Warnings: 0
MySQL localhost:33060+ ssl employee6 SQL> alter table student add constraint fk_course foreign key(course_id) references course(
course_id);
ERROR: 3734: Failed to add the foreign key constraint. Missing column 'course_id' for constraint 'fk_course' in the referenced table
'course'
MySQL localhost:33060+ ssl employee6 SQL> select course_id,count(std_id) as total_students from student group by course_id;
+-----+-----+
| course_id | total_students |
+-----+-----+
| 101       | 3              |
| 102       | 5              |
+-----+-----+

2 rows in set (0.0015 sec)
MySQL localhost:33060+ ssl employee6 SQL> select course_id,count(std_id) as total_students from student group by course_id havin
g count(std_id)>3;
+-----+-----+
| course_id | total_students |
+-----+-----+
| 102       | 5              |
+-----+-----+

2 rows in set (0.0015 sec)
MySQL localhost:33060+ ssl employee6 SQL> select course_id,count(std_id) as total_students from student group by course_id havin
g count(std_id)>3;
+-----+-----+
| course_id | total_students |
+-----+-----+
| 102       | 5              |
+-----+-----+

2 rows in set (0.0015 sec)
MySQL localhost:33060+ ssl employee6 SQL> select course_id,count(std_id) as total_students from student group by course_id havin
g count(std_id)>3 order by total_students desc;
+-----+-----+
| course_id | total_students |
+-----+-----+
| 102       | 5              |
+-----+-----+

1 row in set (0.0012 sec)
MySQL localhost:33060+ ssl employee6 SQL> select course_id,count(std_id) as total_students from student group by course_id havin
g count(std_id)>3 order by total_students desc;
+-----+-----+
| course_id | total_students |
+-----+-----+
| 102       | 5              |
+-----+-----+

1 row in set (0.0013 sec)
MySQL localhost:33060+ ssl employee6 SQL>

```

5. explain why foregin keys are important in realtional databaes?

a)

1. They connect tables - A foreign key links one table to another. Like connecting Students to Courses.
2. They stop bad data* - You can't add a student to a course that doesn't exist. Keeps data clean.
3. They remove duplicate data - Instead of writing course details 100 times, store it once and line

3.Question

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MySQL Shell
1 row in set (0.0013 sec)
MySQL localhost:33060+ ssl employee6 SQL> select user, host from mysql.user;
+-----+-----+
| user          | host      |
+-----+-----+
| gangadhar@localhost | %         |
| mysql.infoschema | localhost |
| mysql.session   | localhost |
| mysql.sys       | localhost |
| root           | localhost |
+-----+-----+

5 rows in set (0.0015 sec)
MySQL localhost:33060+ ssl employee6 SQL> create user 'loki'@'localhost' identified by '123456';
Query OK, 0 rows affected (0.0321 sec)
MySQL localhost:33060+ ssl employee6 SQL> alter user 'loki'@'localhost' identified by '4321';
Query OK, 0 rows affected (0.0213 sec)
MySQL localhost:33060+ ssl employee6 SQL> grant select, insert on school.student to loki@localhost;
ERROR: 1146: Table 'school.student' doesn't exist
MySQL localhost:33060+ ssl employee6 SQL> grant select, insert on 'school.student' to loki@localhost;
ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''school.student' to loki@localhost' at line 1
MySQL localhost:33060+ ssl employee6 SQL> grant select, insert on school.student to loki@localhost;
ERROR: 1146: Table 'school.student' doesn't exist
MySQL localhost:33060+ ssl employee6 SQL> grant select, insert on 'school'.student to 'loki@localhost';
ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''school'.student' to 'loki@localhost' at line 1
MySQL localhost:33060+ ssl employee6 SQL> show databases;
+-----+
| Database |
+-----+
| 96r      |
| demo1    |
| employee |
+-----+

MySQL Shell
MySQL localhost:33060+ ssl employee SQL> CREATE TABLE student (
-> id INT PRIMARY KEY,
-> name VARCHAR(50),
-> class INT
-> );
Query OK, 0 rows affected (0.0384 sec)
MySQL localhost:33060+ ssl employee SQL> GRANT SELECT, INSERT ON employee.student TO 'loki'@'localhost';
Query OK, 0 rows affected (0.0151 sec)
MySQL localhost:33060+ ssl employee SQL> FLUSH PRIVILEGES;
Query OK, 0 rows affected (0.0141 sec)
MySQL localhost:33060+ ssl employee SQL> SHOW GRANTS FOR 'loki'@'localhost';
+-----+
| Grants for loki@localhost |
+-----+
| GRANT USAGE ON *.* TO 'loki'@'localhost' |
| GRANT SELECT, INSERT ON 'employee'.student TO 'loki'@'localhost' |
+-----+

2 rows in set (0.0011 sec)
MySQL localhost:33060+ ssl employee SQL> grant select, insert on 'school'.student to 'loki@localhost';
ERROR: 1064: You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near ''school'.student' to 'loki@localhost' at line 1
MySQL localhost:33060+ ssl employee SQL> grant select, insert on school.student to 'loki@localhost';
ERROR: 1146: Table 'school.student' doesn't exist
MySQL localhost:33060+ ssl employee SQL> REVOKE SELECT, INSERT ON employee.student FROM 'loki'@'localhost';
Query OK, 0 rows affected (0.0152 sec)
MySQL localhost:33060+ ssl employee SQL> show grants for loki@localhost;
+-----+
| Grants for loki@localhost |
+-----+
| GRANT USAGE ON *.* TO 'loki'@'localhost' |
+-----+

1 row in set (0.0006 sec)
MySQL localhost:33060+ ssl employee SQL>
```

5.Explain the meaning and purpose of SSL in database security:

A) SSL = Secure Sockets Layer. Now called TLS. It encrypts the connection between client and database server.

6. Explain the role of DCL commands in sql:

A) Role:

DCL controls who can access the database and what they can do. Used for security. Commands; GRANT, REVOKE